Manual of permanent magnet motor door controller

NSFC01-G220

Ver. 1.5

Preface

Thank you for using NSFC01-G220 door controller!

NSFC01-G220 is a special door controller dedicated to permanent magnet synchronous motor and developed for elevator door system, which integrates door opening and closing logic control and motor drive control. The external system can control the door system only by giving the door opening or closing instructions.

This manual describes the function, characteristics and instructions of the NSFC01-G220 door controller; please read this manual carefully before use (maintenance, inspection, etc.).

Please read and ensure you understand the safety instructions of the product before using it.

Notice
● The illustrations in this manual are for illustration purposes only and may differ slightly from the product you purchased. ● Please understand that due to product upgrades or specification changes, and in order to improve the convenience and accuracy of the instructions, the contents of this manual will be updated in a timely manner without further notice. ● Please contact our company if you need to get the manual again due to damage or loss. ● If you still have some problems in the process of using, please contact our customer service center. ● Service Tel: 400-1120-201

Contents

1 Safety information and precautions

In this manual, the safety precautions are divided into the following two categories:

Danger: Danger due to failure to operate as required, which may lead to personal injury or death.

Warning: Hazards resulting from failure to operate as required, which may result in moderate or minor injury to personnel and damage to equipment.

Please read this chapter carefully when installing, debugging and repairing the controller, and be sure to operate according to the safety precautions required in this chapter. Any injury and loss caused by illegal operation has nothing to do with the company.

1.1 Installation

Danger
■ Please install it on a flame-retardant object such as metal to avoid fire! ■ Keep away from combustible materials to avoid fire!
Warning
■ Do not drop metal conductive objects such as wire heads and screws into the controller to avoid damage to the controller! ■ Please install the controller in a place with less vibration and away from direct sunlight! ■ Please install it in a place that can withstand its weight, so as to avoid personal injury and equipment damage caused by falling! ■ Do not install if the controller is damaged when opening the box! ■ When the packing list is inconsistent with the name of the object, please do not install! ■ Handle with care, otherwise there is a risk of damage to the equipment! ■ Do not touch the components of the controller with your hands, otherwise there is a risk of electrostatic damage!

1.2 Wiring

Danger
■ The instructions in this manual must be followed, and the construction must be carried out by professional electrical engineering personnel to avoid electric shock and injury! ■ The controller and the power supply must be separated by a circuit breaker, otherwise a fire may occur! ■ Please ground the controller correctly according to the standard, otherwise there is a risk of electric shock!

Warning
■ Do not connect the input power supply to the output terminals (U, V, W) of the controller. Pay attention to the marks of the wiring terminals and do not connect the wrong wires, otherwise the controller will be damaged! ■ Make sure that the wiring meets the EMC requirements and the safety standards of the area, otherwise an accident may occur! ■ The communication wire must use the shielded twisted pair with the lay length of 20 ~ 30mm, and the shielding layer shall be grounded! ■ Please make sure that the rated voltage of the product is consistent with the voltage of the AC power supply to avoid injury and fire! ■ Pay attention to check whether there is short circuit in the peripheral circuit connected with the controller and whether the connected circuit is fastened, otherwise the controller will be damaged! ■ Do not carry out withstand voltage test on any part of the controller. This test has been carried out on the product when it leaves the factory. Otherwise, it may cause an accident!

1.3 Power-on

Danger
■ The controller can be powered on only after the cover plate is covered. Do not open the cover plate and touch any input and output terminals of the controller after the controller is powered on, otherwise it may cause electric shock! ■ The wiring of all peripheral accessories must follow the instructions in this manual, and the wiring must be done correctly according to the circuit connection method provided in this manual, otherwise it may cause an accident! ■ Do not change the parameters of the controller manufacturer at will, otherwise the equipment may be damaged! ■ Non-professional technicians are not allowed to detect signals during operation, otherwise personal injury or equipment damage may be caused!

1.4 Maintenance, Inspection and Component Replacement

Warning
■ Please do not repair and maintain the equipment with electricity, otherwise there is a risk of electric shock! ■ Personnel without professional training are not allowed to repair and maintain the controller, otherwise it may cause personal injury or equipment damage! ■ After replacing the controller, the parameters must be set, and all terminals and connectors must be operated in the case of power failure!

- During maintenance and inspection, disconnect the input power supply and wait for 5 minutes, otherwise there is a risk of electric shock.

1.5 Use of external voltage

If the external voltage is not within the rated voltage range specified in the manual, the direct use of the controller is likely to cause damage to the controller device. If necessary, use the appropriate step-up or step-down device for voltage transformation.

1.6 Lightning impulse protection

This series of controllers are equipped with lightning over-current protection devices, which have certain self-protection capability for inductive lightning. However, where lightning occurs frequently, the customer should also install protection at the front end of the controller.

1.7 Altitude and Derating Usage

In the area with an altitude of more than 3000 meters, the heat dissipation effect of the controller becomes poor due to the thin air, so it is necessary to reduce the rating. In this case, please consult our company for technical consultation.

1.8 Scrap disposal

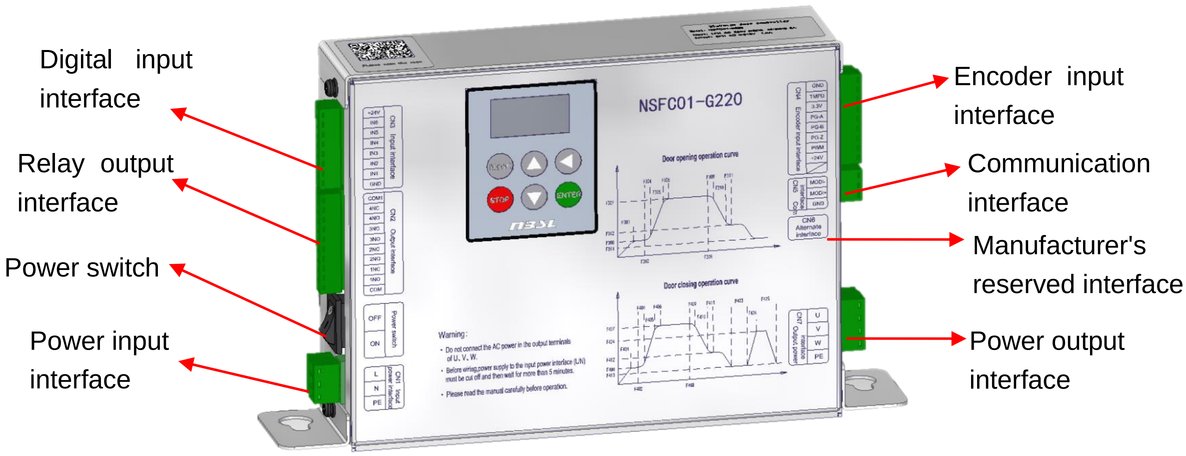
Electrolytic capacitors on the printed board may explode during incineration, and plastic parts will produce toxic gas during incineration, so the controller should be disposed of as industrial waste.

1.9 Adapted motors

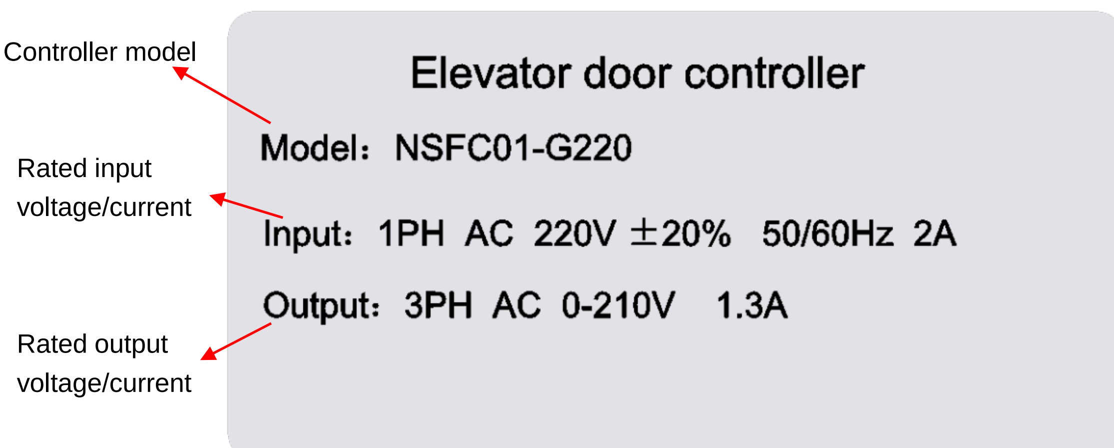
This controller is suitable for AC permanent magnet synchronous motor. Please select the motor according to the nameplate of the controller.

Due to the short circuit inside the cable or motor, the controller will alarm or even be damaged. Therefore, please carry out the insulation short circuit test on the initially installed motor and cable first, and this test should also be carried out frequently in daily maintenance. Note that the controller must be completely disconnected from the part to be tested during this test.

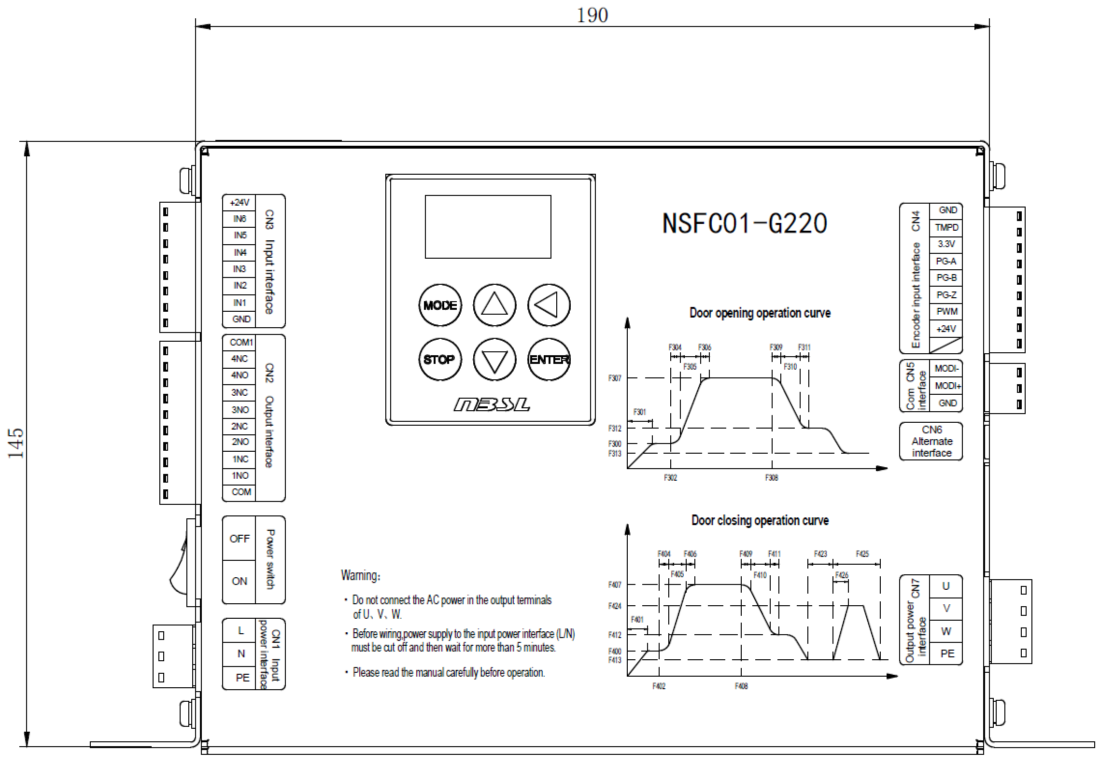
2 Product introduction

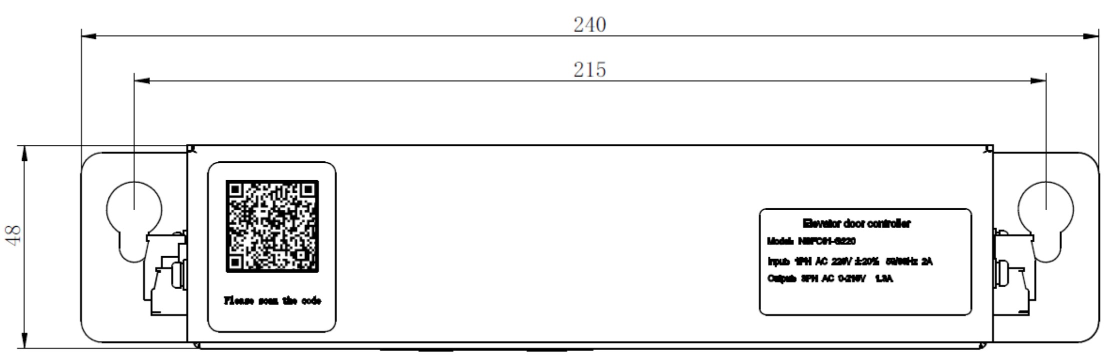


2.1 Nameplate description



2.2 Product dimensions





2.3 Technical specifications

Description	Item	Specifications & Requirement
Control Performance	Maximum output frequency	100Hz
	Method of control	Closed-loop Vector Control of Permanent Magnet Motor
	Speed regulating range	1:1000
	Current resolution	0.01A
	Frequency resolution	0.01Hz
	Steady speed accuracy	±0.1%
	Carrier frequency	2kHz～12kHz
	Overload capacity	150% rated current 60 second
Main function	Motor parameters and magnetic pole position learning function	
	S curve automatic generation function based on distance principle	
	Fault diagnosis function	
	Door width learning function	
	Communication function of upper computer	
	Demonstrate operation function	
	Door opening function	
	Door closing function	
	Low-speed forced door closing function	
	Keeping the door open at the limit position function	
	Keeping the door close at the limit position function	
	Reverse function in case of stall	
	Fault output function	
Protection Function	Support controller overvoltage protection, overheat protection, undervoltage protection, output phase loss protection, etc.	
Environmental Requirements	Degree of IP protection	IP20
	Degree of use protection	Pollution Level：2，Overvoltage level：3
	Temperature of working environment	-10℃~45℃
	Humidity of working environment	When the maximum temperature is 45 ℃, the relative humidity shall not exceed 80%;

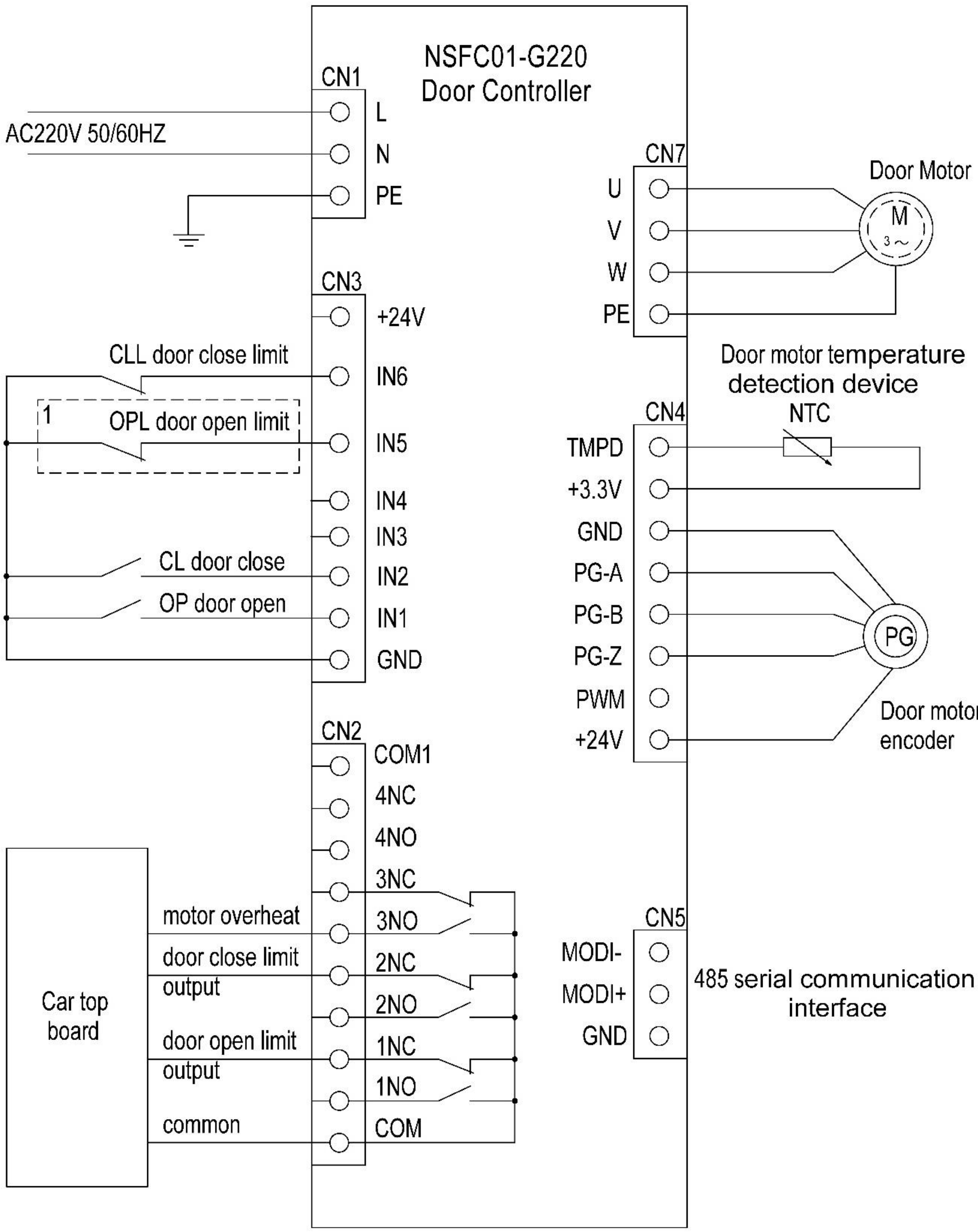
		when the temperature is lower, the humidity may be higher. When the monthly average temperature does not exceed 25 ℃, the monthly average maximum relative humidity shall not exceed 90%, and there shall be no condensation.
	Operating Altitude	Less than 3000m (derating is required for places higher than 3000m)
	Environmental Requirements	No corrosive gas, inflammable gas, etc., not exposed to direct sunlight.
	RoHS Instruction	Compliance with the European Restriction on the Use of Certain Hazardous Substances Directive（RoHS）

3 Electrical installation

3.1 Controller Port Description

Plug-in label	Plug-in name	Terminal designation	Function description
CN1	Power input interface	L/N/PE	1. Single-phase 220V power input 2. PE : grounding terminal
CN2	Relay output interface	1NO/1NC/2NO/2NC 3NO/3NC/COM 4NO/4NC/COM1	1. NO: normally open; NC: normally closed 2. COM: Common Terminal of 1-3NO/NC 3. COM1: Common Terminal of 4NO/NC
CN3	Digital input interface	IN1-IN6/GND/+24V	1. 6-channel digital signal input terminal 2. GND: Common Terminal of IN1-IN6 3. 24V Power supply output
CN4	Encoder input interface	GND	Internal 24V power supply for encoder
		+24V	
		TMPD	Motor over-temperature signal input (analog signal)
		3.3V	
		PG-A	Encoder A Phase
		PG-B	Encoder B Phase
		PG-Z	Encoder Z Phase
		PWM	
CN5	Communication interface	MOD+	Communication Signal
		MOD-	
		GND	Communication grounding
CN6	Interface reserved by the manufacturer		
CN7	Power output interface	U	Motor U phase
		V	Motor V phase
		W	Motor W phase
		PE	Grounding

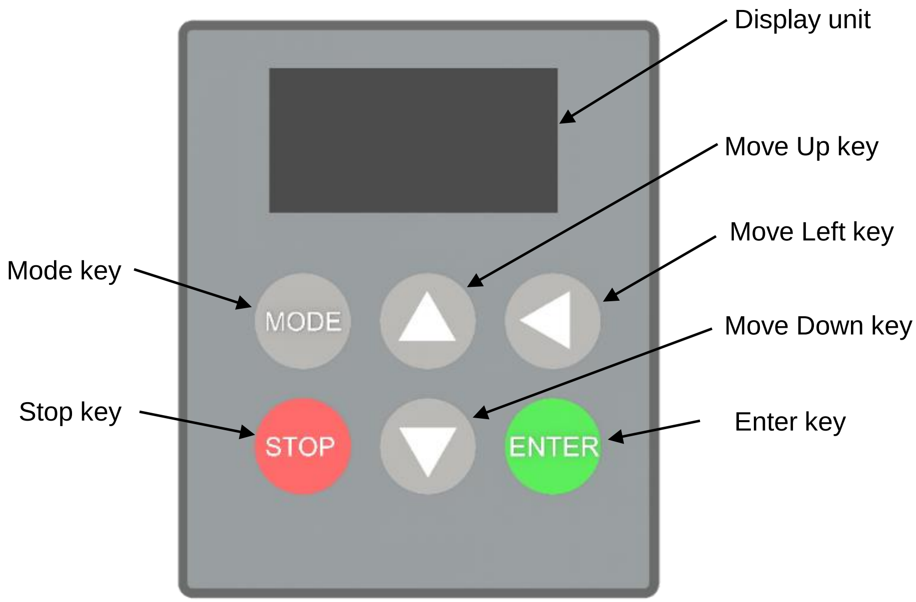
3.2 Electrical wiring diagram



Note 1: The input port shall be connected according to the specific conditions of the site.

4 Operation panel description

4.1 Operation panel layout



4.2 Definition of key functions

Key Name	Description of basic functions
MODE	Mode key: switch to the first-level menu from the main interface and return to the upper-level menu from the lower-level menu
STOP	Stop key: stop the power output to the motor
ENTER	Enter key: enter the lower menu or save the current menu data
▲	Move up key: Increment the menu number or menu data
▼	Move down key: decrement menu number or menu data
◀	Move Left key: switch the display of the main interface and move the flashing position.

4.3 Basic operating instructions

The setting and display of parameters are realized through the integrated operation panel on the door controller, and the operation instructions of each key are as follows:

" **MODE** " Mode key

This key is used for mode switching, and the following operations can be realized by pressing this key:

① Mutual switching between main interface and group menu interface:

main interface $\xrightarrow{\text{MODE}}$ F0 $\xrightarrow{\text{MODE}}$ main interface

② To return from a lower menu to a higher menu:

Submenu No. $\xrightarrow{\text{MODE}}$ Group menu No. such as: F600 $\xrightarrow{\text{MODE}}$ F6

" ENTER " Enter key

This key is used to advance the current display content, mainly in the following cases:

① When the group menu number is currently displayed, press the ENTER key, and the LED displays the submenu corresponding to the group menu:

Group menu No. $\xrightarrow{\text{ENTER}}$ Submenu No. such as: F0 $\xrightarrow{\text{ENTER}}$ F000

② When it is currently displayed as a submenu, press ENTER to enter the menu data:

Submenu No. $\xrightarrow{\text{ENTER}}$ menu data such as:F300 $\xrightarrow{\text{ENTER}}$ 8.00

③ When the menu data is currently displayed, press ENTER to save the menu data and enter the next menu:

menu data $\xrightarrow{\text{ENTER}}$ save menu data such as:F300 $\xrightarrow{\text{ENTER}}$ 8.00 $\xrightarrow{\text{ENTER}}$ F301

" ▲ " Move Up key /" ▼ " Move Down key

This key is used for value adjustment when switching between menus and modifying menu data. The specific description is as follows:

① Switch between group menus: if it is necessary to switch between group menus, enter the group menu and press the ▲ and ▼ keys to switch between the group menus, such as:

F0 $\xrightarrow{\blacktriangle}$ F1 $\xrightarrow{\blacktriangle}$ F2 $\xrightarrow{\blacktriangle}$ F3 ……… $\xrightarrow{\blacktriangle}$ FF

FF $\xrightarrow{\blacktriangledown}$ FE $\xrightarrow{\blacktriangledown}$ FD $\xrightarrow{\blacktriangledown}$ FC ……… $\xrightarrow{\blacktriangledown}$ F0

② Switch between submenus within a group: if you need to switch between submenus, you can enter a group of submenus and press the ▲ and ▼ keys to switch between submenus, such as:

F100 $\xrightarrow{\blacktriangle}$ F101 $\xrightarrow{\blacktriangle}$ F102……… $\xrightarrow{\blacktriangle}$ F115

F115 $\xrightarrow{\blacktriangledown}$ F114 $\xrightarrow{\blacktriangledown}$ F113……… $\xrightarrow{\blacktriangledown}$ F100

③ Adjust the menu data. After entering a certain menu data, use the ▲ and ▼ keys to adjust the menu data:

main interface $\xrightarrow{\text{MODE}}$ F0 $\xrightarrow{\blacktriangle}$ F1 $\xrightarrow{\text{ENTER}}$ F100 $\xrightarrow{\blacktriangle}$ F101 $\xrightarrow{\text{ENTER}}$ 0 2 $\boxed{5}$

$\xrightarrow{\blacktriangle}$ 0 2 $\boxed{6}$ $\xrightarrow{\blacktriangle}$ 0 2 $\boxed{7}$ $\xrightarrow{\blacktriangle}$ 0 2 $\boxed{8}$ $\xrightarrow{\blacktriangledown}$ 0 2 $\boxed{7}$

" ◀ " Move Left key

This key is used to select the data bit to be adjusted, and also used to select different parameters to be monitored in the main menu interface:

① Used to select the data bits to be adjusted:

main interface $\xrightarrow{\text{MODE}}$ F0 $\xrightarrow{▲}$ F1 $\xrightarrow{\text{ENTER}}$ F100 $\xrightarrow{▲}$ F101 $\xrightarrow{\text{ENTER}}$

0 2 $\boxed{5}$ $\xrightarrow{◀}$ 0 $\boxed{2}$ 5 $\xrightarrow{◀}$ $\boxed{0}$ 2 5 $\xrightarrow{◀}$ 0 2 $\boxed{5}$

② The main menu interface is used to select different parameters to be monitored:

Main interface monitoring parameter 1 $\xrightarrow{◀}$ monitoring parameter 2 $\xrightarrow{◀}$ monitoring parameter 3

When the monitoring parameters are displayed on the main interface, the user presses the "ENTER" key, and the LED displays the current parameter serial number.

Use the left shift key to view the list of parameters:

Run Status	Para. No.	Parameter meaning	Remark
Runtime	No.00	Output current (A)	
	No.01	Output Voltage (V)	
	No.02	Operating frequency (Hz)	
	No.03	Busbar voltage (V)	
	No.04	Output Torque (N.M)	
	No.05	Given frequency (Hz)	
	No.06	Actual speed of motor（RPM）	
	No.07	Indication of the door's current position (%)	
	No.08	Current motor temperature (°C)	
	No.09	Input terminal status display	
	No.10	Output terminal status display	
	No.11	Given speed of motor（RPM）	
Stop time	No.0	Software version number	
	No.1	Maximum frequency of door opening（Hz）	
	No.2	Current motor temperature (°C)	
	No.3	Maximum frequency of door closing（Hz）	
	No.4	Indication of the door's current position (%)	
	No.5	Encoder counter values	
	No.6	Encoder pulses per revolution	
	No.7	Busbar voltage (V)	

5 Controller Debugging

Please confirm the following items before powering on:

① The door operator disengages from the landing door without mechanical obstruction;

② All electrical wiring is correct.

5.1 Confirmation of motor parameters

① Input the parameters of the motor in the F1 menu in turn, especially to ensure the correctness of the rated speed of F104 motor and the rated frequency of F105 motor;

② Confirm whether the encoder pulse number is consistent with the data in the F211 menu;

③ Enter menu FA16, manually pull the car door slowly towards the opening direction, observe the display status during the pulling process, and if "OPEN" is displayed, the encoder direction is correct; If "CLOSE" is displayed, adjust the setting value of F212. If the original default value is 0, change it to 1. If the original default value is 1, change it to 0. In this way, the operation direction of the encoder can be switched. Ensure that the rotation towards the door opening direction must be "OPEN".

5.2 Confirmation of connection sequence of motor power line

① Connect the U, V and W wiring terminals of the motor to the U, V and W interfaces of the controller;

② Make sure that the encoder lead on the motor has been correctly connected to the controller terminal and the direction is set correctly;

③ Change the content of menu F001 to 3, and enter the operation mode of ordinary frequency converter;

④ Change the content of menu F005 to 1, press the confirmation key and observe the running direction of the motor;

⑤ If the motor runs in the direction of opening the door, the three-phase power line connection sequence of the motor is correct; If the motor operates in the door closing direction, it indicates that the three-phase power lines of the motor are connected incorrectly in sequence. Press the "STOP" key to stop the operation of the motor, and then exchange the power output connecting lines of U phase and V phase. After the exchange is completed, perform step ④ again until the motor operates in the door opening direction, so as to ensure that the power lines of the motor are connected correctly in sequence.

5.3 Motor magnetic pole position learning

① Change the content of menu F001 to 3;

② No-load magnetic pole position learning mode (the door knife swing arm needs to be lifted to the state of no acting force on the car door)

Change F005 to 2. The motor stops after rotating for a certain distance in the direction of opening the door, and the display shows "0" flashing, and the magnetic pole positioning learning is completed.

③ Loaded pole position learning mode (recommended)

Change F005 to 3, the motor rotates for a certain distance in the direction of opening the door and then stops, the display shows "0" flashing, and the magnetic pole positioning learning is completed. (Note: FA31 is "1.40" version and only supports changing F005 to 2)

5.4 Door width learning

① Correctly configure the limit input signal of the door operator in the F7 menu. If the limit signal is not used, please modify the configuration in the menu to 0 (The default configuration of IN6 in the controller is the door close limit. The door open limit is not configured, additional configuration can be added if necessary);

② Simple door width learning mode: Press the red button "STOP" to ensure that the controller has no power output, then press and hold the up"▲" and down "▼" buttons at the same time, the door controller starts and releases the button, and the controller automatically completes the door width learning. (recommend)

③ Menu door width learning mode: First change the value of menu F001 to 4, and then change the value of menu F003 to 1. After pressing the confirmation key, the controller executes the door width learning operation. The controller drives the door to close first. When the torque reaches the torque set by F602, it turns to open the door. Measure the door width. When the door is opened at limit position, the controller increases the torque to the value set by F602 and then turns to close the door. Stop the voltage output after the door is closed at limit position, and complete the door width learning.

5.5 Demo Running

① After completing the door width learning, change F001 to 2 and enter the demonstration operation mode;

② Set the times of demonstration operation at F608 and F609. F608 and F609 form a number together. F608 is the low digit and F609 is the high digit. Stop operation after the times of demonstration operation reach the set times;

③ Change F002 to 1 and press ENTER to start the demo;

④ In the process of demonstration operation, the dynamic times of operation can be monitored through F610 and F611;

⑤ The F604 and F605 menus are used to set the maintenance time of door open at limit position or door close at limit position during the demonstration operation;

⑥ The F607 menu is used to set whether to automatically start the demo run after power-up.

5.6 Inspection running

① After completing the door width learning, change F001 to 4 and enter the maintenance operation mode;

② Enter the menu F004, press the ▲ key, the controller executes the door opening command, and the controller decelerates immediately after release; press the ▼ key, the controller executes the door closing command, and the controller decelerates immediately after release. The deceleration time is set by the abnormal deceleration time menu F606.

5.7 Terminal signal control running

When the controller is in normal use, it is common to use the terminal to control the opening and closing of the door. The use method is as follows:

① Change the menu data of F001 to 0;

② In the input terminals IN1-IN6 of the controller, select two terminals to access the door opening command OPEN and door closing command CLOSE of the elevator respectively. In the corresponding submenu of group menu F7, the terminal menu data corresponding to the door opening command is set as 01, and the terminal menu data corresponding to the door closing command is set as 02;

③ When the door opening terminal receives a valid command signal, the controller drives the door operator to execute the door opening command; when the door closing terminal receives a valid command signal, the controller drives the door operator to execute the door closing command; if the command is suddenly cancelled during the execution of the command by the controller, the controller will drive the door operator immediately to decelerate to zero, and the deceleration time is set in the F606 menu; When effective command signals appear at both the door opening terminal and the door closing terminal, the door operator executes the door opening command.

5.8 Communication control operation

The controller can communicate with the upper computer through the 485 communication port to realize the control of opening and closing the door. The use method is as follows:

① Confirm that the communication protocol docking has been completed, and connect the CN5 port of the controller to the communication signal of the upper computer;

② Configure the communication baud rate, set the low baud rate in the menu F010, and set the high baud rate in the menu F011;

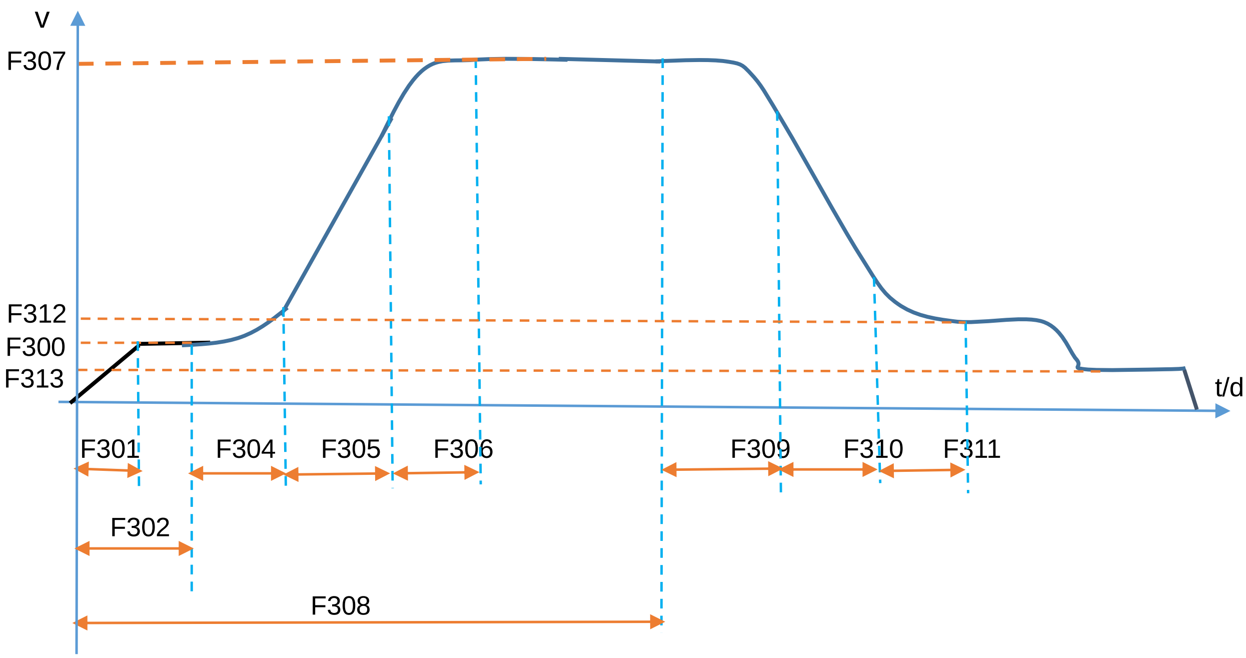
③ Configure the front and rear door settings of the controller. When the F008 data is 0, the controller is used in the front door, and when the F008 data is changed to 1, it is used in the rear door;

④ Change F007 menu to 1, enable 485 communication function, then change F001 to 1, start communication operation mode, power down and then power on to complete the setting, and the controller operates according to the command given by the upper computer.

6 Operation curve of controller

6.1 Operation curve of door opening

Horizontal axis represents time t or distances d Vertical axis represents the velocity v



The door opening process is described as follows:

① After the door opening is started, the controller starts to operate from 0 speed with F303 as the preset torque, accumulates the operation distance from 0 speed, accelerates to the door opening low speed 1 set by F300 after the time set by F301, then operates at the uniform speed of F300, and enters the S curve acceleration operation stage when the accumulated operation distance is greater than or equal to the distance set by F302;

② In the S-curve acceleration operation stage, firstly, accelerating the fillet 1 at the time set by F304, then uniformly accelerating at the time set by F305, accelerating the fillet 2 at the times set by F306 after completing the uniform acceleration time, and reaching the highest operation speed of the controller after accelerating the fillets 2;

③ After that operation is accelerated to the high speed, the controller operates at the highest speed and monitor whether the distance from the complete door closing position as a reference point to the current position point is more than or equal to a deceleration distance set by F308 in real time, if the current operation distance is less than the distance of the deceleration point, the controller continues to operate at the highest speed at a uniform speed, and if the current operation distance is more than or equal to the distance set by the deceleration point, The controller enters a door opening deceleration stage;

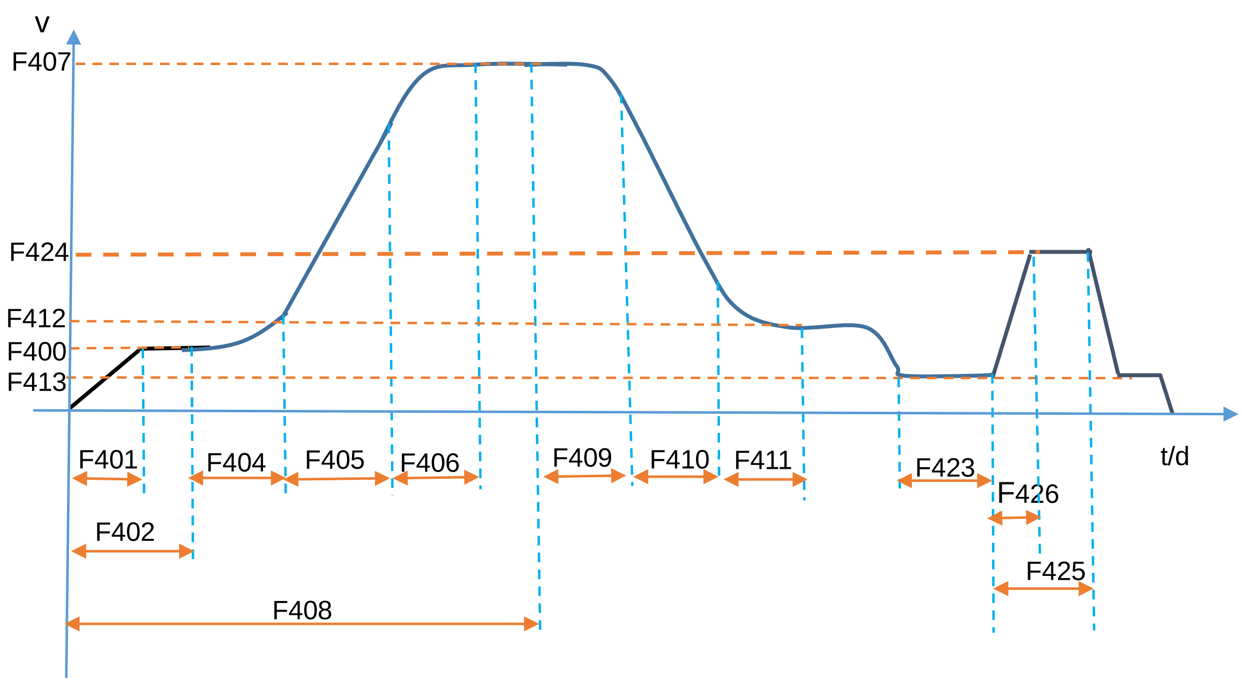
④ In the S-curve deceleration operation stage, firstly operate the deceleration fillet 1 at the time set by F309, then decelerate uniformly at the time set by F310, and then operate to the door opening low speed 2 set by F312 at the time of F311. After the door is opened at limit position, operate at the door opening low speed 3 set in F313 until the speed is reduced to zero after the door is opened and maintained;

⑤ The deceleration point F308 is a parameter of percentage, which is the ratio of the distance from the position when the door is completely closed to the current position of the door to the width of the door;

⑥ The acceleration and deceleration values of the S curve are determined by the speed difference between the high speed and the low speed of the S curve and the time value of the acceleration and deceleration process.

6.2 Operation curve of door closing

Horizontal axis represents time t or distances d　Vertical axis represents the velocity　v



The door closing process is described as follows:

① After the door closing is started, the controller starts to operate from the speed 0, accumulates the operation distance from the speed 0, accelerates to the door closing speed 1 set by F400 after the time set by F401, and then operates at a constant speed of F400, and enters the S curve acceleration operation stage when the accumulated operation distance is greater than or equal to the distance set by F402;

② In the S-curve acceleration operation stage, firstly, accelerating the fillet 1 at the time set by F404, then uniformly accelerating at the time set by F405, accelerating the fillet 2 at the times set by F406 after completing the uniform acceleration time, and finally reaching the highest operation speed of the controller after accelerating the fillets 2;

③ After that operation is accelerated to the high speed, the controller operates at the highest speed and monitor whether the distance from the completely door-opening position as a reference point to the current position point is more than or equal to a deceleration distance set by F408 in real time, if the current operation distance is less than the distance of a deceleration point, the controller continues to operate at the highest speed at a uniform speed, and if the current operation distance is more than or equal to the distance set by the deceleration point, The controller enters a door closing deceleration stage;

④ In the S-curve deceleration operation stage, firstly, the deceleration fillet 1 is operated at the time set by F409, then the uniform deceleration is carried out at the time set by F410, and then the operation is carried out to the door closing speed 2 set by F412 at the time of F411, and after the door is closed at limit position, the operation is carried out at the door closing speed 3 set in F413 for the time set in F423. After the time set for closing the door at low speed 3, the door controller operates at the time set by F426 to the speed set by F424, and operates at a constant speed for a time of [F425]- [F426], and then decelerates to the opening speed at low speed 3 set by F413 until the speed is reduced to zero after closing the door;

⑤ The deceleration point F408 is a parameter of percentage, which is the ratio of the distance from the reference zero point when the door is fully opened to the current position of the door to the width of the door;

⑥ The acceleration and deceleration values of the S curve are determined by the speed difference between the high speed and the low speed of the S curve and the time value of the acceleration and deceleration process.

7 Menu parameters

Func. Group	Menu No.	Menu Name	Parameter Range	Unit	Default Value	Modify Type
F0 Operating parameter	F000	Standby			0	1
	F001	Controller Operation Mode	0：Terminal Input Control Operation Mode 1：485 communication control operation mode 2：Auto demo run mode 3：Normal Inverter Operation Mode 4: Door width learning and manual door opening and closing mode		0	1
	F002	Start command of door opening and closing operation during controller commissioning	1：Start the test run		0	1
	F003	Door width learning instruction	0: No operation; 1: Start door width learning		0	1
	F004	Manual door opening and closing command	Manual door opening and closing command input When F001 is set to 4: Press the ▲ key to open the door Press the ▼ button and close the door		0	
	F005	Open-loop operation and magnetic pole position learning of motor	When F001 is set to 3: 0: No operation 1: Open Loop Run Startup 2. No-load rotating magnetic pole positioning learning 3. On-load closed-loop magnetic pole positioning learning (recommended)		0	1
	F006	Output Voltage Percentage for Open Loop Operation	50.0-120.0%	% *Rated voltage	100.0%	1
	F007	485 communication port enabled	0：Turn off 485 communication		0	1

Func. Group	Menu No.	Menu Name	Parameter Range	Unit	Default Value	Modify Type
			1：Open 485 communication			
	F008	485 Front and rear door setting during communication	0：Controller for front door 1：Controller for rear door		0	0
	F009	Magnetic pole position learning current values	0.4-3.0	0.1* rated current	1.0	1
	F010	485 Low value of communication baud rate	0000-9999	bps	8400	0
	F011	485 High value of communication baud rate	0000-0012	10kbps	3	0
F1 Basic parameters of motor	F100	Type of motor	00: Synchronous motor		0	1
	F101	Rated power of motor	000-250	W	50	1
	F102	Rated voltage of motor	000-270	V	50	1
	F103	Rated current of motor	0.20—4.00	A	1.10	1
	F104	Rated speed of motor	1—3000	rpm	180	1
	F105	Rated frequency of motor	1.00—99.99	Hz	24.00	1
	F106	Rated back electromotive force of motor	1—250	V	45	1
	F107	Rated torque of motor	0.00—15.00	N.M	2.70	1
	F108	D-axis inductance Ld of synchronous motor	0.1—999.9	mH	10.5	1
	F109	Q-axis inductance Lq of synchronous motor	0.1—999.9	mH	10.5	1
	F110	Motor stator phase resistance Rs	0.01—99.99	Ω	84.00	1
	F111	Magnetic pole position of synchronous motor encoder	0000—2048		329	1
	F112	Absolute position angle value of encoder when Z point appears	0000-359.9	°	0	1
	F113	Motor real-time angle	0000-359.9	°	0.7	1
	F114	Threshold value of motor temperature protection	80-120	°C	100	

Func. Group	Menu No.	Menu Name	Parameter Range	Unit	Default Value	Modify Type
F2 Motor control parameter	F200	Proportional gain parameter P1 of velocity loop	0—128		25	0
	F201	Velocity loop integration time parameter T1	0.01—15.00	S	0.50	0
	F202	Proportional gain parameter P2 of velocity loop	0—128		20	0
	F203	Velocity loop integration time parameter T2	0.01—15.00	S	0.50	0
	F204	Parameter switching frequency F1	0.00—F105	Hz	3.00	0
	F205	Parameter switching frequency F2	F204—F105	Hz	8.00	0
	F206	Velocity feedback filter coefficient	0—99		10	0
	F207	Proportional gain of current loop	1—512		80	0
	F208	Current loop integral gain	1—512		30	0
	F209	Current feedback filter coefficient	0—64		10	0
	F210	Standby			0.2	0
	F211	Pulses per revolution of encoder	1—9999		1024	1
	F212	Encoder pulse direction	0—1		1	1
	F213	PWM carrier frequency selection	2—12	kHz	10	0
F3 Door opening control parameter	F300	Open the door and start at low speed 1	0.01—15.00	Hz	2.50	0
	F301	Open door start acceleration time	0.1—5.0	S	0.4	0
	F302	Low-speed running distance of opening door	0.1—40.0	%* Door width	5.0	0
	F303	Opening starting moment	0.1—200.0	%* Rated torque	30.0	1

Func. Group	Menu No.	Menu Name	Parameter Range	Unit	Default Value	Modify Type
F3 Door opening control parameter	F304	Open door acceleration fillet 1 time toa1	0.1—9.9	S	0.2	0
	F305	Uniform acceleration time for opening door toa2	0.1—9.9	S	0.3	0
	F306	Open door acceleration fillet 2 time toa3	0.1—9.9	S	0.3	0
	F307	Maximum speed of opening door	0.01—F105	Hz	18.00	0
	F308	Distance of door opening deceleration point	30.0—90.0	%* Door width	63.5	0
	F309	Open door deceleration fillet 1 time tod1	0.1-9.9	S	0.3	0
	F310	Uniform deceleration time for opening door tod2	0.1-9.9	S	0.3	0
	F311	Open door deceleration fillet 2 time tod3	0.1-9.9	S	0.3	0
	F312	Open the door at low speed 2	0.01—10.00	Hz	2.50	0
	F313	Low speed 3 after door open at the limit position	0.01—5.00	Hz	1.0	0
	F314	Slow opening speed	0.01—F307	Hz	5.00	0
	F315	Reference moment of door open at limit position	0.1—150.0	% * Rated torque	90.0	0
	F316	Output delay time of door open at limit position	0.0—9.9	S	1	0
	F317	Maintain torque of door open at limit position	0.1—100.0	% * Rated torque	60.0	0
	F318	Blocked moment of opening door	0.1—150.0	% * Rated torque	130.0	0

Func. Group	Menu No.	Menu Name	Parameter Range	Unit	Default Value	Modify Type
	F319	Judgment time of blocked door opening	0.01—9.99	S	4.00	0
	F320	Reference point of door open at limit position	80.0～99.9	% * Door width	97.0	0
	F321	Output reference point of door open at limit position signal	00.0—99.9	% * Door width	97.5	0
	F322	Torque maintenance time after door open at limit position	0.0—999.9	S	999.9	0
	F323	Opening time limit	000.0—999.9	S	000.0	0
	F324	Limit percentage of detection torque of door opening jam	1.0-150.0	%* Rated torque	80	0
	F325	Standby				
	F326	Door opening function type selection	When F627 = 2, Bit0 is valid. When the door running to the open limit position for the first time after power on, Bit0 = 0: Output door open limit Bit 0 = 1: Do not output door open limit		0	0
F4 Door closing control parameter	F400	Close door start low speed 1	0.01—9.99	Hz	3.00	0
	F401	Close door start acceleration time	0.1—5.0	S	0.3	0
	F402	Close door low speed 1 running distance	0.1—40.0	%* Door width	4.0	0
	F403	Standby				0
	F404	Close door acceleration fillet 1 time tca1	0.1—9.9	S	0.3	0
	F405	Closed door uniform acceleration time tca2	0.1—9.9	S	0.4	0
	F406	Close door acceleration fillet 2 time tca3	0.1—9.9	S	0.2	0
	F407	Maximum closing speed	0.01—F105	Hz	16.00	0

Func. Group	Menu No.	Menu Name	Parameter Range	Unit	Default Value	Modify Type
F4 Door closing control parameter	F408	Distance of door closing deceleration point	10.0—90.0	%* Door width	67.5	0
	F409	Close door deceleration fillet 1 time tcd1	0.1—9.9	S	0.2	0
	F410	Closure uniform deceleration time tcd2	0.1—9.9	S	0.2	0
	F411	Close door deceleration fillet 2 time tcd3	0.1—9.9	S	0.4	0
	F412	Close door low speed 2	0.01—9.99	Hz	1.60	0
	F413	Low speed 3 after door close at limit position	0.01—9.99	Hz	0.80	0
	F414	Slow closing speed	0.01—F407	Hz	4.00	0
	F415	Reference moment of door close at limit position	0.1—150.0	% * Rated torque	90.0%	0
	F416	Output delay time of door close at limit position	0.0—9.9	S	1.5	0
	F417	Maintain torque of door close at limit position	0.1—150.0	% * Rated torque	40.0	0
	F418	Closure blocking moment	0.1—150.0	% * Rated torque	110.0	1
	F419	Judgment time of closed door obstruction	0.01—9.99	S	3.00	0
	F420	Torque maintenance time after door close at limit position	0.0—999.9	S	999.9	0
	F421	Reference point of door close at limit position	0.0—99.9	% * Door width	96.0	0
	F422	Reference point of door close limit signal output	0.0—99.9	% * Door width	98.0	0

Func. Group	Menu No.	Menu Name	Parameter Range	Unit	Default Value	Modify Type
	F423	Close door low speed 3 running time	0.00—9.99	S	1.00	0
	F424	Synchronous closing speed of door cutter	0.01—F105	Hz	2.00	0
	F425	Synchronous closing time of door cutter	0.00—9.99	S	0.70	0
	F426	Acceleration time of synchronous door cutter closing operation	0.01—9.99	S	0.50	0
	F427	Closure 150N blocking moment	0.1—150.0	% * Rated torque	100.0	1
	F428	Determination time of blocking moment of closing door 150N	0.01—9.99	S	3.0	0
	F429	The controller 150N is blocked and enabled by reverse door opening	0: No 150N blocked automatic reverse door opening 1: Allow 150N blocked automatic reverse door opening		0	0
	F430	Closing time limit	000.0—999.9	S	000.0	0
	F431	Percentage of torque limit detected by door closing jam	1.0-150.0	% * Rated torque	50	0
	F432	Door closing function type selection	Detects the movement distance of the door knife and outputs the door close limit signal: 0: close 1: enable 2: detects close command		1	0
F5 Distance parameter	F500	Low bit of gate width pulse count	0000—9999		——	1
	F501	High bit of gate width pulse count	0000—9999		——	1
	F502	Door open limit switch pulse position low	0000—9999		3280	1
	F503	Door open limit switch pulse position high	0000—9999		0002	1

Func. Group	Menu No.	Menu Name	Parameter Range	Unit	Default Value	Modify Type
F5 Distance parameter	F504	Door close limit switch pulse position coordinates	0000—9999		704	1
	F505	Allowable error range of door knife action distance	1—99.9	mm	40	0
	F506	Input port filter time	0000-9999	ms		0
	F507	Door width learning automatically generates running curve	When door width learning completed: 0:automatically generates running curve 1:do not automatically generates running curve		1	0
F6 Door operating parameters	F600	Door width learning speed	1.00——15.00	Hz	5.00	0
	F601	Maintain torque after blocking detection in power-on positioning operation	1.0-150.0	% * Rated torque	70	
	F602	Reference moment of opening in place when learning door width	0.1—150.0	% * Rated torque	120.0	0
	F603	Locating run torque on power-on	0.1—150.0	% * Rated torque	100.0	0
	F604	The maintenance time of demonstrate running when door open at limit position	0.5—500.0	S	2.0	0
	F605	The maintenance time of demonstrate running when door close at limit position	0.5—500.0	S	2.0	0
	F606	Abnormal deceleration time of door opening and closing	0.1—5.0	S	0.2	0
	F607	Control function type selection	Bit0 Demo Run Boot Mode Bit0=0: Manual start Bit0=1: Automatic start		0	0

Func. Group	Menu No.	Menu Name	Parameter Range	Unit	Default Value	Modify Type
			Bit1 enable limit switch update position coordinate Bit1=0: open Bit1=1: close			
	F608	Demonstrate the low number of run settings	0000—9999	Times	1000	0
	F609	High number of demonstration run settings	0000—9999	Ten thousand times	9999	0
	F610	Demonstrate that the actual number of runs is low	0000—9999	Times	0	0
	F611	Demonstrate that the actual number of runs is high	0000—9999	Ten thousand times	0	0
	F612	Operating frequency setting of common frequency converter	0 -- F105	0.01 Hz	6	1
	F613	Restore factory parameters	0: No function 169: Restore factory value		0	1
	F614	Selection of control parameters of different portal cranes	000: Do not load parameters 001: Load Group 001 parameters 002: Load Group 002 parameters 003: Load Group 003 parameters ... 100: Load the 100th set of parameters Total of 100 sets of parameters can be selected		0	1
	F615	Software update enabled	Implement software update after modifying to 6746 + enter		0	0
F7 Input terminal	F700	Standby	The input terminal function is defined as follows 00: This terminal is not used 01: Open door command input (normally open)		0	1
	F701	IN1 input terminal function selection			1	1

Func. Group	Menu No.	Menu Name	Parameter Range	Unit	Default Value	Modify Type
function configuration	F702	IN2 input terminal function selection	02: Close door command input (normally open) 03: Door open limit signal input (normally closed) 04: Door close limit signal input (normally closed) 05: Light screen signal input (normally open) 06: Slow door closing command input (normally open) 07: Door open limit signal input (normally open) 08: Door close limit signal input (normally open) 09: Fire signal input (normally on) 10: Safety touch panel input (normally closed) 11: Safety touch panel input (normally open) 12: Light screen signal input (normally closed) 13: Door lock signal input (normally open) 14: Door lock signal input (normally closed) 15: Motor temperature is too high input (normally on)输入		2	1
	F703	IN3 input terminal function selection			6	1
	F704	IN4 input terminal function selection			5	1
	F705	IN5 input terminal function selection			13	1
	F706	IN6 input terminal function selection			8	
F8 Output terminal function configuration	F800	Door width percentage setting output of door closing process	0.0%—100.0%	0.1%* Door width	20.0%	1
	F801	Relay 1NC/NO output function selection	Output terminal functions are defined as follows: 00: No output 01: Door open limit output 02: Door close limit output 03: Fault signal output 04: Blocked signal output		1	1
	F802	Relay 2NC/NO output function selection			2	1
	F803	Relay 3NC/NO output function selection			9	1

Func. Group	Menu No.	Menu Name	Parameter Range	Unit	Default Value	Modify Type
	F804	Relay 4NC/NO output function selection	05: Door lock output 06: Reopen the door output 07: Motor overtemperature signal output 08: Light curtain/safety shoe action output 09: Output of closing percentage (shielding safety touch panel light curtain) 10: Fault and motor overtemperature signal output at the same time 11: Fault signal inverted output 12: Motor overtemperature signal inverted output 13: Fault and motor overtemperature signal inverted output at the same time 14: Blocked signal inverted output		4	1
F9	F900	Current fault type	Display mode: EXY.Z XY denotes fault type, XY=00 denotes no fault, Z denotes fault subcode, and Z=0 denotes no subcode The fault types are defined as follows: E01.0: IPM Module Overcurrent Fault E02.0: Accelerated overvoltage fault E02.1: Deceleration overvoltage fault E02.2: Uniform Velocity Overvoltage Fault E03.0: Undervoltage fault E04.0: Accelerated overcurrent fault E04.1: Deceleration overcurrent fault		0	2
	F901	Type of first failure			0	2
	F902	Type of second fault			0	2
	F903	Third fault type			0	2
	F904	The 4th fault type			0	2
	F905	The fifth fault type			0	2
	F906	The 6th fault type			0	2
	F907	The 7th fault type			0	2
	F908	The 8th fault type			0	2
	F909	The 9th fault type			0	2
	F910	Tenth fault type			0	2

Func. Group	Menu No.	Menu Name	Parameter Range	Unit	Default Value	Modify Type
Fault Information Query	F911	The 11th fault type	E04.2: Uniform overcurrent fault E05.0: Motor overheating fault E06.0: Module Overheat Fault E07.0: Motor Overload Fault E08.0: Reserved E09.0: Motor overspeed fault E10.0: Excessive speed deviation fault E11.0: Parameter memory failure E12.0: Reserved E13.0: Reserved E14.0: Reserved E15.0: Door close limit switch failure E16.0: Door width learning failure E17.0: Output phase failure E18.1: Encoder reverse E18.2: Encoder Z signal abnormal failure E18.3: Encoder A/B signal abnormal or door jam failure E19.0: Stationary magnetic pole position learning failure E19.1: Rotating magnetic pole position learning failure E19.2: Verification error of magnetic pole position when opening the door for the first time after power on E20.0: Reserved E21.0: 485 Communication Failure		0	2
	F912	Twelfth fault type			0	2
	F913	The 13th fault type			0	2
	F914	The 14th fault type			0	2
F9 Fault Information Query	F915	The 15th fault type			0	2
	F916	Running state at the last failure	Opening and closing status, speed process		0	2

Func. Group	Menu No.	Menu Name	Parameter Range	Unit	Default Value	Modify Type
	F917	Output current at the last fault			0	2
	F918	Operating frequency at the last failure			0	2
	F919	Bus voltage at the last fault			0	2
	F920	Output torque at the last fault				
	F921	Input and output status at the last failure	The high two-bit LED indicates the input state, and the low two-bit LED indicates the output state		0	2
	F922				0	
	F923	Fault clearing instruction	Change to 55. Press enter to clear the fault record		0	1
FA Monitoring parameter	FA00	Given current	0.00—9.99	A	0	2
	FA01	Feedback current	0.00—9.99	A	0	2
	FA02	Given frequency	0.00—99.99	Hz	0	2
	FA03	Feedback frequency	0.00—99.99	Hz	0	2
	FA04	Output torque value	0.00-9.00	N.M	0	2
	FA05	Ratio of output torque to rated torque	0.0-200.0	% * Rated torque	0	2
	FA06	Output voltage	0-250	V	0	2
	FA07	Bus voltage	0-400.0	V	0	2
	FA08	Count value of encoder	0-9999		0	2
	FA09	Encoder Z point count indication	0-9		0	2
	FA10	Current motor pole angle	0-359.9	°	0	2
	FA11				0	2
	FA12	Motor temperature	0—150	℃	0	2
	FA13	Portal crane position (%)	0.0—100.0	% * Door width	0	2
	FA14	Low gate position pulse	0—9999		0	2
	FA15	Gate position pulse high position	0—9999		0	2

Func. Group	Menu No.	Menu Name	Parameter Range	Unit	Default Value	Modify Type
FA Monitoring parameter	FA16	Running direction of motor	OPEN,CLOS		0	2
	FA17	Terminal input door opening and closing command indication	OPEN,CLOS		0	2
	FA18	Input terminal indication	/		0	2
	FA19	Output terminal indication	/		0	2
	FA20	Cumulative power-up time low bit	0000—9999	min	0	2
	FA21	Cumulative power-up time high bit	0000—9999	h	0	2
	FA22	Cumulative running time low bit	0000—9999	min	0	2
	FA23	Cumulative running time high bit	0000—9999	h	0	2
	FA24	Door opening and closing times low bit	0000—9999	Times	0	2
	FA25	Door opening and closing times high bit	0000—9999	10k Times	0	2
	FA26	Door open limit input signal indication	OPL,0		0	2
	FA27	Door close limit input signal indication	CLL,0		0	2
	FA28	Actual speed of motor	0000-9999	rpm	0	2
	FA29	Given speed of motor	0000-9999	rpm	0	2
	FA30	485 receive data status indication	Increment in case of normal communication and unchanged in case of failure		0	2
	FA31	Software version number	00.00-99.99		0	2
	FB01	Standby				
	FB02	Record F614 preset value				

Note 1: If the modify type is 0, it can be modified in real time; if it is 1, it is not allowed to be modified during operation; if it is 2, it is not allowed to be modified.

8 Menu Description

8.1 F0 group Operating parameter

Menu No.	Menu Name	Function description
F000	Standby	
F001	Controller operation mode	0: Terminal input control operation mode 1: 485 communication data control operation mode, which generally communicates with the car roof 2: Automatically demonstrate the running mode, F002=1+ENTER starts running 3: Common frequency converter operation mode, which is used to check whether the relevant wiring (encoder, power line) of the controller is correct 4. Door width learning and manual door opening and closing; When learning the door width, start through F003 menu; Manual door opening and closing is realized by pressing the "▲" and "" buttons after entering the F004 menu
F002	Open and close the door to run the start command when the controller is debugged	Set to 01, the controller starts the test operation in the direction of closing the door. After closing in place, it automatically opens the door and runs circularly
F003	Door width learning start	After all the preliminary work is completed, start the door width learning as follows: 1. Change menu F001 to 4; 2. Change F003 to 1, press the confirmation key, and the controller enters the door width learning state. First, the controller runs in the door closing direction. After receiving the door close limit signal and the torque reaches the specified value (F602), the controller changes the operation direction and runs in the door opening direction to enter the door width measurement state. After the door opening is blocked and the torque reaches the specified value (F602), the controller reverses the operation direction. Operate toward the door closing direction, complete the door width learning

		after the door is closed at limit position, and the controller stops outputting; 3. If terminal control mode is adopted, change F001 to 0; if communication control mode is adopted, change F001 to 1; Change F001 to 2 if Demo Operating Mode is used
F004	Manual door opening and closing command	First modify the menu F001=4, then enter this menu and press the "▲" button to open the door; Press the "" button to close the door
F005	1. Open-loop operation and pole position learning	1. When F001=03 and F005=01, the controller outputs the torque set by F006 to supply power to the motor. If the motor runs slowly in the direction of opening the door, the UVW wiring is correct; On the contrary, it is necessary to adjust any two-phase line sequence of UVW. 2. When F001=03 and F005=02, the controller outputs the torque set by F006 to supply power to the motor, and the motor runs slowly in the direction of opening the door. After learning the Z signal of the encoder, it stops, and automatically saves the magnetic pole position data to the F111 menu. 3. When F001=03 and F005=03, the closed loop of the controller runs in the direction of opening the door, stops after learning the Z-point signal of the encoder, and automatically saves the magnetic pole position to the F111 menu.
F006	Output Voltage Percentage for Open Loop Operation	Set according to the percentage of the rated voltage of the motor. If the door panel is heavy and can't be pulled in open loop operation, the percentage value can be appropriately increased
F007	485 communication port enabled	The 485 communication port has two purposes: ①The upper computer sends the door opening and closing operation command to the door controller by means of communication. In this mode, please complete the setting according to the following steps: 1. Change the F001 menu to 1 to enter the communication mode; 2. Change F007 menu to 1 and start 485 communication;

		3. If there are two door controllers, be sure to set the value of unit F008 to distinguish the front and rear doors; 4. Set the communication baud rate. The two units F010 and F011 together are the communication baud rate. After the baud rate is set, it needs to be powered off and then powered on again to work. ②The IOT platform collects the running status of the gantry crane controller in real time. In this mode, please complete the setting according to the following steps: 1. Change menu F001 to 0 and enable the terminal control mode; 2. Change F007 menu to 1 and start 485 communication; 3. If there are two door controllers, be sure to set the value of unit F008 to distinguish the front and rear doors; 4. Set the communication baud rate. The two units F010 and F011 together are the communication baud rate. After the baud rate is set, it needs to be powered off and then powered on to work.
F008	485 Front and rear door setting during communication	When there are two controllers at the front and rear doors, the two controllers are connected to the upper computer through one bus. After receiving the message from the upper computer, the door controller extracts the identifier to identify the command flow from the upper computer. If the identifier is consistent with that set in F008, the received command will be executed, otherwise it will be discarded; The host computer also identifies which controller the received message comes from through the identifier.
F009	Magnetic pole position learning current values	During magnetic pole learning, the controller injects the current set by F009 unit into the motor to make the magnetic circuit of the motor enter the saturation state. The controller infers the magnetic pole position by detecting the current values of different directions. The user does not need to modify the value of this unit. If necessary, it needs to be modified under the guidance of the manufacturer's engineer.

F010	485 Low value of communication baud rate	The communication baud rate is composed of the values of the two menus, and its calculation method is: baudrate = (F011) × 10000 + (F010). After changing the value of the communication baud rate each time, it needs to be powered off once to work.
F011	485 High value of communication baud rate	

8.2 F1 group Basic parameters of motor

Menu No.	Menu Name	Function description
F100	Type of motor	0: Synchronous motors, currently only synchronous motors are supported 1: Asynchronous motor, standby
F101	Rated power of motor	Power value of door drive motor in W
F102	Rated voltage of motor	Rated voltage value of door drive motor in V
F103	Rated current of motor	Rated current value of door drive motor in A
F104	Rated speed of motor	Rated speed of door drive motor in RPM
F105	Rated frequency of motor	Rated frequency value of door drive motor in Hz
F106	Rated back electromotive force of motor	Back EMF value of door drive motor in V
F107	Rated torque of motor	Output torque of door drive motor at rated current, unit is N.M
F108	D-axis inductance Ld of synchronous motor	Excitation inductance value of door drive motor in mH
F109	Q-axis inductance Lq of synchronous motor	Quadrature axis inductance value of door drive motor in mH
F110	Motor stator phase resistance Rs	Stator resistance value of gate drive motor in Ω
F111	Magnetic pole position of synchronous motor encoder	Minimum Pulse Difference of Magnetic Pole of Synchronous Motor Relative to Encoder Zero
F112	Absolute position angle value of encoder when Z point appears	The value of the pole position angle, expressed in PWM, that is locked when the Z point occurs
F113	Motor real-time angle	Magnetic pole position indication of door motor
F114	Threshold value of motor temperature protection	When the system detects that the internal temperature of the motor reaches this set temperature, the system gives shutdown protection.

8.3 F2 group Motor control parameter

Menu No.	Menu Name	Function description
F200	Proportional gain parameter P1 of velocity loop	When the motor operating frequency is less than F204, the proportional parameters used in the controller speed loop, the smaller the value, the smaller the proportional gain, the larger the value, the larger the proportional gain, the higher the value to a certain extent will cause the motor operating low-speed oscillation, the specific value is determined according to the field situation
F201	Velocity loop integration time parameter T1	When the motor operating frequency is less than F204, the integral time parameter used in the controller speed loop, the smaller the value, the faster the speed response, and the larger the value, the slower the speed response. If the value is reduced to a certain extent, the motor will run at a low speed. The specific value is determined according to the field situation
F202	Proportional gain parameter P2 of velocity loop	When the motor operating frequency is greater than F205, the proportional parameter used in the controller speed loop, the smaller the value, the smaller the proportional gain, the larger the value, the larger the proportional gain, the larger the value to a certain extent will cause the motor operating oscillation, the specific value is determined according to the field situation
F203	Velocity loop integration time parameter T2	When the motor operating frequency is greater than F205, the integral time parameter used by the controller speed loop, the smaller the value, the faster the speed response, the larger the value, the slower the speed response, the smaller the value to a certain extent will cause the motor operating oscillation, the specific value is determined according to the field situation
F204	Parameter switching frequency F1	This parameter is used to determine the working range of F200 and F201 parameters. From 0.00 Hz to the frequency range set by this parameter, F200 and F201 parameters work
F205	Parameter switching frequency	This parameter is used to determine the working range

Menu No.	Menu Name	Function description
	F2	of F202 and F203 parameters, and the F202 and F203 parameters work when the frequency range is larger than the frequency range set by this parameter
F206	Velocity feedback filter coefficient	This parameter is used for filtering calculation of feedback speed. It is recommended that users use it according to the default value and use this parameter carefully
F207	Proportional gain of current loop	It is used to adjust the proportional gain coefficient of the current loop. The larger the value, the faster the proportional response of the current loop. When the parameter is increased to a certain extent, the motor will give out a whistling sound, which can be adjusted by the user according to the actual situation on site
F208	Current loop integral gain	It is used to adjust the integral gain coefficient of the current loop. The larger the value, the faster the integral response of the current loop. When the parameter increases to a certain extent, the motor will oscillate. Users can adjust it according to the actual situation on site
F209	Current feedback filter coefficient	This parameter is used to filter the current value sampled from ADC. If the parameter is too small, it may produce a certain whistling sound during the operation of the motor, and if the value is too large, it will easily cause the fluctuation of the speed of the motor
F211	Pulses per revolution of encoder	According to the parameter setting of the encoder on the tail shaft of the motor, if the set parameter does not conform to the pulse number of the actual encoder, the motor will not be able to rotate or the torque will not be generated after several turns of rotation
F212	Encoder pulse direction	It is used for adjusting the growth direction of the encoder counter. If the encoder pulse counter is incremented when 0 is set, the encoder pulse counter is decremented when 1 is set, and vice versa
F213	PWM carrier frequency selection	This parameter is used to set the carrier frequency of svpwm. If the carrier frequency is less than 4KHz, there will be obvious current noise. If the carrier frequency is

Menu No.	Menu Name	Function description
		greater than 12KHz, the switching loss and heat loss of the module will increase

8.4 F3 group Door opening control parameter

Menu No.	Menu Name	Function description
F300	Open the door and start at low speed 1	Running speed when opening the door, the door crane starts from zero after receiving the running command, accelerates to this speed after the time set by F301, drives the door to run and opens the lock hook, so that the door knife hangs the door ball
F301	Open door start acceleration time	Used to adjust the transition time from zero speed acceleration to low speed 1 when opening the door
F302	Low-speed running distance of opening door	It is used to set the distance for the door crane to run at a low speed of opening the door, and ensure that the door knife has been smoothly hung on the door ball within this distance. When this value is too small, the door knife will hit the door ball at a higher speed to produce noise. When this value is too large, it will affect the whole opening time. The value is determined according to the needs of the site
F303	Opening starting moment	It is used to set the preset torque for opening the door. Because the elevator door has a clamping force when it is closed, adjusting this parameter to an appropriate value can accelerate the starting process of opening the door and improve the efficiency
F304	Open door acceleration fillet 1 time toa1	The excessive fillet time from zero-speed start-up to uniform acceleration of S curve; Toa1+toa2+toa3 is the opening acceleration time
F305	Uniform acceleration time for opening door toa2	Uniform acceleration time of S curve; Toa1+toa2+toa3 is the opening acceleration time
F306	Open door acceleration fillet 2 time toa3	The excessive fillet time from uniform acceleration to maximum speed when the S curve rises; Toa1+toa2+toa3 is the opening acceleration time
F307	Maximum speed of opening door	It is used to set the maximum speed of opening the door, and the maximum speed setting and the time

Menu No.	Menu Name	Function description
		setting of opening the door determine the acceleration value of opening the door. The maximum speed value should conform to the principle, m is the mass of the door, and v is the maximum speed of opening the door$0.5 \times \mathrm{m}v^2 \leq 10$
F308	Distance of door opening deceleration point	This parameter is used to set the position of the door opening deceleration point. After exceeding this position, the door crane starts to decelerate. The reference zero of this position is the position when the door is completely closed; This distance parameter value is a percentage of the ratio of the distance value from the reference zero point to the deceleration point to the door width distance
F309	Open door deceleration fillet 1 time tod1	The excessive fillet time of S curve from maximum speed to uniform deceleration; Tod1+tod2+tod3 is the deceleration time of opening door
F310	Uniform deceleration time for opening door tod2	Uniform deceleration time of S curve; Tod1+tod2+tod3 is the deceleration time of opening door
F311	Open door deceleration fillet 2 time tod3	The transition time of S curve from uniform deceleration to low opening speed 2; Tod1+tod2+tod3 is the deceleration time of opening door
F312	Open the door at low speed 2	The portal crane decelerates from the highest speed through the S curve and reaches this speed, and then runs at this speed at a constant speed until the door is opened at limit position
F313	Low speed 3 after door open at the limit position	After door opened at limit position, switch to this speed, and the torque increases to the opening maintenance torque and then enters the maintenance state
F314	Slow opening speed	When the door crane is powered on for the first time or there is a slow door opening command, the door crane runs at this speed
F315	Reference moment of door open at limit position	In the case of not using the door open limit switch, door opened at limit position is considered to be reached after the obstructed torque reaches this set torque after the door opening deceleration is completed

Menu No.	Menu Name	Function description
F316	Output delay time of door open at limit position	After the door is opened at limit position, this parameter is used to adjust the lag time of the door open limit signal output from the relay
F317	Maintain torque of door open at limit position	A moment applied in the direction of opening the door after the door is opened at limit position
F318	Blocked moment of opening door	In the process of opening the door, when the output torque of the controller exceeds the value set in this menu and reaches the time set in F319, it is determined that the opening of the door is blocked
F319	Judgment time of blocked door opening	After the opening moment reaches the value set by F318, it is considered that the opening is blocked after continuing the time value set by this menu
F320	Reference point of door open at limit position	This menu value is a coordinate value. During the process of opening the door, if the position of the door exceeds this value, it will be considered that the door is opened at limit position
F321	Output reference point of door open at limit position signal	Coordinate position point of door open limit signal output
F322	Torque maintenance time after door open at limit position	Setting the duration of the moment when door open at limit position, if the unit is set to 999.9 seconds, it will be maintained all the time. If it is not equal to 999.9 seconds, it will be maintained according to the set time
F323	Opening time limit	Limit the response time of the open door command, and do not respond if it timeout. Set to 0, there is no time limit.
F324	Limit percentage of detection torque of door opening jam	After the system detects the door opening jamming, the output current is adjusted to the set value of F324.
F325	Standby	
F326	Door opening function type selection	This parameter is represented by hexadecimal converted from binary, and each bit of binary represents a function. Bit0 is the output enable function of the door open in the first run after power-on.When F627 = 2, Bit0 is valid. When the door running to the open limit position for the first time after power on, Bit0 = 0: Output door open limit Bit 0 = 1: Do not output door open limit

8.5 F4 group Door closing control parameter

Menu No.	Menu Name	Function description
F400	Close door start low speed 1	The running speed when the door is closed, the door crane starts from zero after receiving the door closing operation command, and accelerates to this speed after the time set by F401
F401	Close door start acceleration time	Used to adjust the time from zero speed acceleration to low speed 1 when closing the door
F402	Low speed running distance of door closing	It is used to set the running distance of the door crane at the closing speed of 1, and ensure that the door knife has smoothly hung the door ball in the closing direction within this distance
F404	Close door acceleration fillet 1 time tca1	The excessive fillet time from zero-speed start-up to uniform acceleration of S curve; Tca1+tca2+tca3 is the closing acceleration time
F405	Closed door uniform acceleration time tca2	Uniform acceleration time of S curve; Tca1+tca2+tca3 is the closing acceleration time
F406	Close door acceleration fillet 2 time tca3	The excessive fillet time from uniform acceleration to maximum speed when the S curve rises; Tca1+tca2+tca3 is the closing acceleration time
F407	Maximum closing speed	It is used to set the maximum speed for closing the door. The maximum speed setting and the time setting for closing the door determine the acceleration value of closing the door. The maximum speed value should conform to the principle, m is the mass of the door, and v is the maximum speed of closing the door $0.5 \times \mathrm{m}v^2 \leq 10J$
F408	Distance of door closing deceleration point	This parameter is used to set the position of the door closing deceleration point. After exceeding this position, the door crane starts to decelerate. The reference zero of this position is the position when the door is fully opened in place; This distance parameter value is a percentage of the ratio of the distance value from the reference zero point to the deceleration point to the door width distance
F409	Close door deceleration fillet 1 time tcd1	The excessive fillet time of S curve from maximum speed to uniform deceleration; TCD1 + TCD2 + TCD3

Menu No.	Menu Name	Function description
		is the closing deceleration time.
F410	Closure uniform deceleration time tcd2	Uniform deceleration time of S curve; TCD1 + TCD2 + TCD3 is the closing deceleration time
F411	Close door deceleration fillet 2 time tcd3	Transition time of S curve from uniform deceleration to closing low speed 2; TCD1 + TCD2 + TCD3 is the closing deceleration time
F412	Close door low speed 2	The gantry crane decelerates from the highest speed through the S curve and reaches this speed, and then runs at this speed at a constant speed until the door is closed at limit position
F413	Low speed 3 after door close at limit position	After the door is closed at limit position, switch to this speed and run at a constant speed
F414	Slow closing speed	When the door crane is powered on for the first time or there is a slow door closing command, the door crane will close at this speed
F415	Reference moment of door close at limit position	In the case of not using the door close limit switch, the door closed at limit position is considered to be reached after the blocked torque reaches this set torque value after the door deceleration is completed
F416	Output delay time of door close at limit position	After door closed at limit position, use this parameter to adjust the lag time of outputting the door close limit signal from the relay
F417	Maintain torque of door close at limit position	A moment applied in the closing direction to close the door after the door is closed at limit position
F418	Closure blocking moment	During the closing process, when the output torque of the controller exceeds the value set in this menu and reaches the time set in F419, it is judged that the closing is blocked. After the closing is blocked, the controller opens the door in reverse
F419	Judgment time of closed door obstruction	After the closing moment reaches the value set by F418, it is considered that the closing is blocked after continuing the time value set by this menu
F420	Torque maintenance time after door close at limit position	Set the door close limit moment holding time, if the unit is set to 999.9 seconds, it will be maintained all the time. If it is not equal to 999.9 seconds, it will be maintained according to the set time. After reaching

Menu No.	Menu Name	Function description
		the maintenance time, the controller will stop the power output
F421	Reference point of door close at limit position	This menu value is a coordinate value. When the door is closed, the door is considered to be in place if the position of the door exceeds this value
F422	Reference point of door close limit signal output	Coordinate position point of door close limit signal output
F423	Close door low speed 3 running time	This parameter is only applicable to the case of synchronous door cutter. After the speed of the controller is switched to three stages of low opening speed, the closing operation of synchronous door cutter is implemented after the time set in this menu
F424	Synchronous closing speed of door cutter	Running speed value of synchronous door cutter
F425	Synchronous closing time of door cutter	This time includes the acceleration time F426 and the uniform speed running time of the synchronous door cutter
F426	Acceleration time of synchronous door cutter closing operation	This time is the transition time from closing the door at low speed 3 to F424
F427	Closure 150N blocking moment	During the closing process, when the output torque of the controller exceeds the value set in this menu and reaches the time set by F428, it is judged that the closing of 150N is blocked
F428	Determination time of blocking moment of closing door 150N	After the closing moment reaches the value set by F427, it is considered that the closing is blocked after continuing the time value set by this menu
F429	The controller 150N is blocked and enabled by reverse door opening	0: No 150N blocked automatic reverse door opening 1: Allow 150N blocked automatic reverse door opening
F430	Closing time limit	Limit the response time of the close door command, and do not respond if it timeout. Set to 0, there is no time limit.
F431	Percentage of torque limit detected by door closing jam	After the system detects the door closing jamming, the output current is adjusted to the set value of F431.
F432	Door closing function type selection	Detects the movement distance of the door knife and outputs the door close limit signal function:

Menu No.	Menu Name	Function description
		0: close 1: enable 2: detects close command

8.6 F5 group Distance parameter

Menu No.	Menu Name	Function description
F500	Low bit of gate width pulse count	The door width is the distance from when the door is fully closed to when the door is fully open. This distance is represented by the encoder pulse count. This menu is the lower order of the count. Each unit represents one encoder pulse.
F501	High bit of gate width pulse count	Each unit represents ten thousand encoder pulses
F502	Door open limit switch pulse position low	If the door open limit switch is used, this value represents the lower 4 digits of the pulse count value from the time when the door is completely closed to the moment when the door open limit switch is operated
F503	Door open limit switch pulse position high	The high value of the door open limit switch, each unit represents 10,000 encoder pulses
F504	Door close limit switch pulse position coordinates	If the door close limit switch is used, this value represents the pulse count value from the time when the door is completely closed to the time when the door close limit switch leaves
F505	Allowable error range of door knife action distance	The allowable error of the actual distance of the door running during each operation of the door knife
F506	Input port filter time	Each unit represents 1ms
F507	Door width learning automatically generates running curve	When door width learning completed: 0:automatically generates running curve 1:do not automatically generates running curve

8.7 F6 group Door operating parameters

Menu No.	Menu Name	Function description
F600	Door width learning speed	Running speed value in the process of door width learning
F601	Maintain torque after blocking	After the system is powered on, after the door jamming

Menu No.	Menu Name	Function description
	detection in power-on positioning operation	is detected in the process of positioning the opening and closing curve, the current is reduced to the set value of this menu.
F602	Reference moment of opening in place when learning door width	When the door width learning is blocked, when the torque value set in this menu is reached, it is considered that the door is opened at limit position and the door width measurement is completed
F603	Locating run torque on power-on	Maximum torque output limit for the first operation after power-on
F604	The maintenance time of demonstrate running when door open at limit position	The time required to wait for the demonstrate runtime to switch from door open limit to closing the door
F605	The maintenance time of demonstrate running when door close at limit position	The time required to wait for the demonstrate runtime to switch from door close limit to opening the door
F606	Abnormal deceleration time of door opening and closing	In the process of opening and closing the door, if the current given instruction disappears, the controller decelerates from the current running speed to zero at the time set in this menu
F607	Control function type selection	This parameter is represented by hexadecimal converted from binary, and each bit of binary represents a function. Bit0 Demo Run Boot Mode Bit0=0: Manual start Bit0=1: Automatic start Bit1 enable limit switch update position coordinate Bit1=0: open Bit1=1: close
F608	Demonstrate the low number of run settings	The lower number of times required to run in demonstration mode, each numerical unit represents one time
F609	High number of demonstration run settings	The high number of times required to run in demonstration mode, and each numerical unit represents ten thousand times
F610	Demonstrate that the actual number of runs is low	The lower number of actual runs in demonstration mode, and each numerical unit represents one time
F611	Demonstrate that the actual number of runs is high	The high number of actual runs in demonstration mode, and each numerical unit represents ten thousand times

Menu No.	Menu Name	Function description
F612	Running speed of ordinary frequency converter	The speed setting value for startup runtime when F001 is set to 3 mode

8.8 F7 group Input terminal function configuration

Menu No.	Menu Name	Function description
F700	Standby	
F701	Function selection of IN1 input terminal	00: This terminal is not used 01: Open door command input (normally open) 02: Close door command input (normally open) 03: Door open limit signal input (normally closed) 04: Door close limit signal input (normally closed) 05: Light screen signal input (normally open) 06: Slow door closing command input (normally open) 07: Door open limit signal input (normally open) 08: Door close limit signal input (normally open) 09: Fire signal input (normally on) 10: Safety touch panel input (normally closed) 11: Safety touch panel input (normally open) 12: Light screen signal input (normally closed) 13: Door lock signal input (normally open) 14: Door lock signal input (normally closed) 15: Motor temperature is too high input (normally on)
F702	Function selection of IN2 input terminal	
F703	Function selection of IN3 input terminal	
F704	Function selection of IN4 input terminal	
F705	Function selection of IN5 input terminal	
F706	Function selection of IN6 input terminal	

8.9 F8 group Output terminal function configuration

Menu No.	Menu Name	Function description
F800	Door width percentage setting output of door closing process	After the door closing process reaches this set value, the F804 is configured as 09 output contact signal to cooperate with the control cabinet to shield the light curtain or safety touch panel.
F801	Relay 1NC/NO output function selection	00: No output 01: Door open limit output

F802	Relay 2NC/NO output function selection	02: Door close limit output 03: Fault signal output 04: Blocked signal output 05: Door lock output 06: Reopen the door output 07: Motor overtemperature signal output 08: Light curtain/safety shoe action output 09: Output of closing percentage (shielding safety touch panel light curtain) 10: Fault and motor overtemperature signal output at the same time 11: Fault signal inverted output 12: Motor overtemperature signal inverted output 13: Fault and motor overtemperature signal inverted output at the same time 14: Blocked signal inverted output
F803	Relay 3NC/NO output function selection	
F804	Relay 4NC/NO output function selection	

8.10 F9 group Fault Information Query

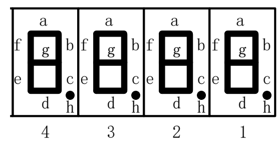
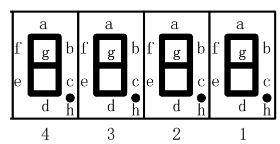
Menu No.	Menu Name	Function description
F900	Current fault type	XY denotes fault type, Z denotes fault subcode, XY=00 denotes no fault, and Z=0 denotes no subcode The fault types are defined as follows: E01.0: IPM module overcurrent E02.0: Acceleration overvoltage E02.1: Deceleration overvoltage E02.2: Uniform Velocity Overvoltage E03.0: Undervoltage E04.0: Acceleration overcurrent E04.1: Deceleration overcurrent E04.2: Uniform Velocity overcurrent E05.0: Motor overheating E06.0: Module overheat E07.0: Motor overload E08.0: Reserved E09.0: Motor overspeed E10.0: Excessive speed deviation
F901	Type of fault No.01	
F902	The 02nd fault type	
F903	Type of fault No.03	
F904	The 04th fault type	
F905	The 05th fault type	
F906	The 06th fault type	
F907	Type of fault No.07	
F908	The 08 th fault type	
F909	The 09th fault type	
F910	Tenth fault type	

Menu No.	Menu Name	Function description
F911	The 11th fault type	E11.0: Parameter memory failure E12.0: Reserved E13.0: Reserved E14.0: Reserved E15.0: Door close limit switch failure E16.0: Door width learning failure E17.0: Output phase failure E18.1: Encoder reverse E18.2: Encoder Z signal abnormal failure E18.3: Encoder A/B signal abnormal or door jam failure E19.0: Stationary magnetic pole position learning failure E19.1: Rotating magnetic pole position learning failure E19.2: Verification error of magnetic pole position when opening the door for the first time after power on E20.0: Reserved E21.0: 485 Communication Failure
F912	Twelfth fault type	
F913	The 13th fault type	
F914	The 14th fault type	
F915	The 15th fault type	
F916	Running state at the last failure	Including door opening and closing status and speed process
F917	Output current at the last fault	Unit: A
F918	Operating frequency at the last failure	Unit: Hz
F919	Bus voltage at the last fault	Unit: V
F920	Output torque at the last fault	Unit: N.M
F921	Input and output status at the last failure	The upper two LEDs indicate the input status and the lower two LEDs indicate the output status
F923	Fault information clearing instruction	Change this menu to 55. Press enter and clear all fault records

8.11 FA group Monitoring parameter

Menu No.	Menu Name	Function description
FA00	Given current	Used to view the vector composite value of excitation current and torque current, unit: A
FA01	Feedback current	Used to view the real-time current value output from

		the controller to the motor, unit: A
FA02	Given frequency	Instantaneous given operating frequency value of controller in Hz
FA03	Feedback frequency	Instantaneous actual operating frequency value of the controller in Hz
FA04	Output torque value	Instantaneous output torque value of motor, unit: N.M
FA05	Ratio of output torque to rated torque	(TO/TR) × 100. 0% To: Output torque value, Tr: Rated torque value
FA06	Output voltage	Single-phase voltage output to motor terminal
FA07	Bus voltage	DC voltage value of three-phase alternating current after rectification
FA08	Count value of encoder	The dynamic value of the encoder pulse counter, which is used to observe the increase and decrease direction of the encoder
FA09	Encoder Z point count indication	The counter is incremented every time the Z-point occurs, which is used to indicate the number of occurrences of the Z-signal
FA10	Current motor pole angle	Dynamically indicate the current magnetic pole position angle
FA12	Motor temperature	The real-time temperature of the motor is calculated according to the temperature sensor on the motor
FA13	Portal crane position (%)	Taking the position of the door when it is completely closed as the reference zero point, the distance difference between the current position and the zero point is divided by the value of the door width multiplied by 100
FA14	Low gate position pulse	Low bit of the gate position pulse count value. The gate position pulse count value refers to the distance from the position where the gate is completely closed as 0 to the current position in terms of pulses. The calculation formula is (FA15) × 10000 + (FA14).
FA15	Gate position pulse high position	High bit of the gate position pulse count value.
FA16	Motor running direction indication	According to the motor running direction indication determined by the encoder direction, OPEN; is displayed when the encoder increases; Display CLOS when encoder decrements
FA17	Terminal input door opening	According to the direction indication of door opening and

	and closing command indication	closing determined by the port input command, OPEN is displayed when the door opening command is input on the input terminal; Display CLOS when the door command is input on the input terminal
FA18	Input terminal indication	8 segments of LED are corresponding to IN1-IN6 of input terminals, and the corresponding LED display segment lights up when the input of corresponding terminals is valid  The a-f of LED No.1 correspond to IN1-IN6
FA19	Output terminal indication	Use 8 segments of LED and output terminal TC1-TC4 to correspond, the corresponding terminal output is effective when the corresponding LED display segment lights up.  The a-d of LED No.1 correspond to 1NO/NC-4NO/NC
FA20	Low cumulative power-on time	The lower part of the running time from power-on to now, in minutes
FA21	High cumulative power-on time	High level of running time from power-on to now, in hours
FA22	Low cumulative run time	The lower part of the cumulative value of power output time from the beginning of power-on to the present, in minutes
FA23	High cumulative power-on time	The high position of the cumulative value of power output time from power-on to now, in hours
FA24	The number of opening and closing doors is low	Low number of running times of portal crane from power-on to now
FA25	The number of opening and closing doors is high	The high number of running times of portal crane from the beginning of power-on to the present
FA26	Door open limit input signal indication	If the door open limit signal is used, OPL will be displayed every time the effective input of the door open limit signal is detected, and 0 will be displayed when the effective input of the door opening signal is not

		detected
FA27	Door close limit input signal indication	If the door close limit signal is used, CLL will be displayed every time a valid input of the door close limit signal is detected, and 0 will be displayed when no valid input of the door-closing signal is detected
FA28	Actual speed of motor	Displays the current actual instantaneous speed of the motor in RPM (rpm)
FA29	Given speed of motor	Displays the current given instantaneous speed of the motor in RPM (rpm)
FA30	485 Communication status indication	The menu data is incremented when the communication is successful, and remains unchanged when the communication fails.
FA31	Version number of software	Displays the current software version
FB01	Standby	
FB02	Record F614 preset value	

Appendices A Fault Information and Troubleshooting

Fault Code	Fault Type Definition	Possible Causes	Solution	Remark
E01.0	IPM module overcurrent	1.Motor phase line short circuit to ground 2.Drive module failure 3.wrong magnetic pole angle	1. Troubleshoot circuit and motor faults 2. Replace the inverter 3. Check the encoder and relearn the magnetic pole angle	Stop and lock after three automatic resets
E02.0	Acceleration overvoltage	1. Acceleration slope fillet time is too short 2. Input voltage is too high 3. Inverter fault	1. Modify acceleration curve parameters 2. Check the power supply, adjust the voltage to the normal range 3. Replace the inverter	Stop and lock after three automatic resets
E02.1	Deceleration overvoltage	1. Input voltage is too high 2. Deceleration slope fillet time is too short 3. Inverter fault	1. Check the power supply, adjust the voltage to the normal range 2. Adjust the deceleration curve 3. Replace the inverter	Stop and lock after three automatic resets
E02.2	Uniform Velocity Overvoltage	1. Input voltage is too high 2. Inverter fault	1. Check the power supply, adjust the voltage to the normal range 2. Replace the inverter	Stop and lock after three automatic resets
E03.0	Undervoltage	1. Input voltage is too low 2. Inverter fault 3. Poor contact power line or plug-in	1. Check the power supply, adjust the voltage to the normal range 2. Replace the inverter 3. Check the power line and plug-in tightness	Automatic reset after the voltage is normal
E04.0	Acceleration overcurrent	1.Wrong magnetic pole angle 2. Overload 3.Motor phase line short circuit to ground	1.Check the encoder and relearn the magnetic pole angle 2.Check for mechanical resistance and clear obstacles 3. Troubleshoot circuit and motor faults	Stop and lock after three automatic resets
E04.1	Deceleration overcurrent	1.Wrong magnetic pole angle 2. Overload	1.Check the encoder and relearn the magnetic pole angle	Stop and lock after three automatic

		3.Motor phase line short circuit to ground	2.Check for mechanical resistance and clear obstacles 3. Troubleshoot circuit and motor faults	resets
E04.2	Uniform Velocity overcurrent	1.Wrong magnetic pole angle 2. Overload 3.Motor phase line short circuit to ground	1.Check the encoder and relearn the magnetic pole angle 2.Check for mechanical resistance and clear obstacles 3. Troubleshoot circuit and motor faults	Stop and lock after three automatic resets
E05.0	Motor overheating	1. Overload 2.Inverter F114 is set too small 3.High temperature	1.Check for mechanical resistance and clear obstacles 2. Adjust the temperature detection value of F114 3.Ventilation & heat dissipation	Error prompt, but can continue to run
E06.0	Module overheat	1.High temperature 2. Overload	1.Ventilation & heat dissipation 2.Check for mechanical resistance and clear obstacles	Automatic reset after the temperature is normal
E07.0	Motor overload	1. Overload 2.Inverter overload detection parameters are unreasonable	1.Check for mechanical resistance and clear obstacles 2.Adjust the inverter detection overload parameter or increase the current value of F103	Automatic reset after the load is normal
E08.0	Reserved			
E09.0	Motor overspeed f	1.Wrong magnetic pole angle 2.Acceleration and deceleration slope fillet time is too short	1.Check the encoder and relearn the magnetic pole angle 2.Adjust the acceleration and deceleration curve	Stop and lock after three automatic resets
E10.0	Excessive speed deviation	1. Overload 2. Encoder exception	1.Check for mechanical resistance and clear obstacles 2.Check the encoder and relearn the magnetic pole	Stop and lock after three automatic resets

			angle	
E11.0	Parameter memory failure	1.Inverter data read and write abnormal	1. Replace the inverter	Stop and lock
E12.0	Reserved			
E13.0	Reserved			
E14.0	Reserved			
E15.0	Door close limit switch failure	1.Limit switch signal error	1. Troubleshoot switch failures 2. Exclude the door closing position switch circuit	Stop and lock
E16.0	Door width learning failure	1.The door operator is blocked when studying 2. Door close switch fault 3.Door close limit signal wiring error	1.Check for mechanical resistance and clear obstacles 2. Check the door closing signal 3. Check wiring	Commissioning failure
E17.0	Output phase loss	1. The motor UVW output wire is loose or the plug-in is not inserted tightly 2. Inverter fault 3. Motor failure	1. Check wiring and plug-in tightness 2. Replace the inverter 3. Replace the motor	Stop and lock
E18.1	Encoder reverse	1.The encoder direction is inconsistent with the motor UVW running direction	1. Adjust any two-phase line sequence of the motor 2. Modify F212 parameters	Commissioning failure
E18.2	Encoder Z signal abnormal failure	1. Encoder wiring error 2 .Encoder damaged	1. Check encoder wiring 2. Replace the encoder	Stop and lock after one door opening and closing cycle
E18.3	Encoder A/B signal abnormal or door jam failure	1. Encoder wiring error 2.Door machine mechanical jam	1. Check encoder wiring 2. Replace the encoder 3.Check for mechanical resistance and clear obstacles	1.Encoder failure unable to operate 2.Automatic recovery after jam remove
E19.0	Stationary magnetic pole position learning failure	1. Motor failure 2. Encoder failure	1. Check encoder 2. Check the three-phase windings of motor	Commissioning failure

E19.1	Rotating magnetic pole position learning failure	1. Motor failure 2. Encoder failure	1. Check encoder 2. Check the three-phase windings of motor	Commissioning failure
E19.2	Verification error of magnetic pole position when opening the door for the first time after power on	1. Motor failure 2. Encoder failure	1. Check encoder 2. Check the three-phase windings of motor	Stop and lock after three automatic resets
E20.0	Reserved			
E21.0	485 Communication Failure	1.Communication line plug loose 2. Inverter communication port fault	1. Check wiring and plug-in tightness 2. Replace the inverter 3. Check the communication port of the host computer	Communication failure

Appendices B　Version Change Record

Date of issue	Version No.	Content of change	Remark
Aug. 2021	**V1.0**	Issue of the first edition	Software version:V1.51
Sep. 2021	**V1.1**	5.3 add a magnetic pole position learning mode (F005 to 3), menu update F114 the temperature control, and update the progress output function of the door closing process.	Software version:V1.51
Oct. 2021	**V1.2**	5.8 communication control operation is added for commissioning, F007-011 communication function setting parameter is added for menu, and FA30 fault description is added.	Software version:V1.51
Mar. 2022	**V1.3**	Appendices A added fault analysis and countermeasures, and fault & temperature detection output logic.	Software version:V1.51
May. 2022	**V1.4**	Menu added FA18,19 terminal instructions, and added the magnetic pole position 18.3 fault code.	Software version:V1.58
Jun. 2022	**V1.5**	5.4 added simple door width learning mode,added F507,when door width learning completed, automatically generates running curve, add F432 door close limit detection door closing command output mode.	Software version:V1.59